# Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass

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## Abstract

*Traditional multi-view 3D reconstruction methods operate pairwise, requiring O(N²) matching that limits scalability and accumulates errors as view counts grow. We present Fast3R, a transformer-based approach that performs 3D reconstruction of 1000+ images in a single forward pass by jointly reasoning over all views simultaneously. Our method replaces pairwise processing with a fusion transformer using all-to-all attention over patches from every view, augmented with image-index positional embeddings and position interpolation to enable "train short, test long" capability. Fast3R predicts both local and global pointmaps with confidences for all images at once, eliminating the need for costly global alignment. Experimental results demonstrate state-of-the-art pose accuracy on CO3Dv2 (RRA@15=99.7%, mAA(30)=82.5%) with 251.1 FPS—over 300× faster than DUSt3R. Our approach scales to 1000+ views while DUSt3R fails beyond 48 views, achieving competitive reconstruction quality with orders-of-magnitude speedups.*

## 1. Introduction

Multi-view 3D reconstruction from unposed images is a fundamental problem in computer vision with applications ranging from robotics to augmented reality. Traditional structure-from-motion pipelines [13] rely on feature matching, pose estimation, and dense reconstruction stages that operate sequentially and require careful initialization. Recent end-to-end methods like DUSt3R [18] and MASt3R [7] have shown promising results by directly regressing pointmaps from image pairs, but they still operate pairwise, requiring O(N²) comparisons and costly global alignment that becomes computationally prohibitive as the number of views N increases.

The core limitation of pairwise approaches is their inability to leverage global context during initial prediction. Each pair is processed independently, leading to accumulation of errors and inconsistencies that must be resolved through expensive optimization. This fundamental bottleneck limits scalability—existing methods fail to handle more than a few dozen views efficiently and struggle with memory constraints.

We propose Fast3R, a transformer-based architecture that addresses these limitations by jointly reasoning over all views in a single forward pass. Our key insight is that a transformer with all-to-all attention can directly predict multi-view pointmaps while maintaining global consistency, eliminating the need for pairwise processing and subsequent global alignment. Fast3R ingests N unordered, unposed images and outputs local and global pointmaps with confidences for all views simultaneously.

Our main contributions are:

1. A single-pass multi-view pointmap regressor with all-to-all attention that eliminates O(N²) pairwise scaling and global alignment requirements.
2. Image-index positional embeddings with position interpolation that enable generalization from 20 training views to 1000+ inference views.
3. A dual-head design predicting both local and global pointmaps with confidence-weighted supervision, where local-to-global alignment improves reconstruction fidelity.
4. State-of-the-art pose accuracy with unprecedented throughput: 251.1 FPS with competitive reconstruction quality, enabling 1000+ view inference while existing methods fail beyond 48 views.

Extensive experiments demonstrate that Fast3R achieves state-of-the-art performance on camera pose estimation (RRA@15=99.7% on CO3Dv2) while being orders of magnitude faster than existing approaches, opening new possibilities for large-scale 3D reconstruction applications.

## 2. Related Work

**Traditional Multi-view Reconstruction.** Classical structure-from-motion (SfM) methods [13] follow a sequential pipeline: feature extraction and matching, pose estimation, and dense reconstruction. While these methods have achieved remarkable success, they require careful initialization, are sensitive to outliers, and scale poorly with the number of views due to the quadratic complexity of pairwise matching. Bundle adjustment optimization becomes increasingly expensive as the number of views grows, often requiring careful outlier filtering and robust optimization techniques.

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**Learning-based Reconstruction.** Recent works have explored end-to-end learning approaches to address traditional SfM limitations. DUSt3R [18] introduced the paradigm of directly regressing pointmaps from image pairs using transformer architectures, eliminating the need for explicit feature matching and camera calibration. MASt3R [7] extended this approach with dense matching capabilities, while Spann3R [15] introduced spatial memory to maintain global consistency. However, these methods still operate pairwise and require global alignment optimization through gradient descent or closed-form solutions, which becomes computationally prohibitive for large view sets. The global alignment step in DUSt3R requires solving a least-squares problem for camera poses, while MASt3R employs iterative optimization that can take minutes for moderate view counts. Both approaches suffer from error accumulation as inconsistencies from pairwise predictions compound during alignment.

**Transformer Architectures for 3D Vision.** Vision transformers [5] have shown remarkable success in various computer vision tasks. Their ability to model long-range dependencies through self-attention makes them particularly suitable for multi-view reasoning. Recent works have explored transformers for depth estimation [10], stereo matching, and point cloud processing. The success of attention mechanisms in capturing global context has motivated their application to 3D reconstruction, where understanding relationships between distant views is crucial for accurate geometry estimation.

**Scalability in 3D Reconstruction.** Scaling 3D reconstruction to large numbers of views has been a persistent challenge across both traditional and learning-based methods. Classical approaches suffer from quadratic complexity in feature matching and bundle adjustment optimization that becomes prohibitively expensive. Recent learning methods face memory constraints due to pairwise processing requirements. Some works have explored hierarchical reconstruction strategies, incremental approaches, or distributed optimization, but these typically sacrifice either accuracy or global consistency for computational efficiency.

**Efficiency and Real-time Reconstruction.** Computational efficiency in multi-view reconstruction has gained increasing attention for practical applications. Traditional real-time systems often limit the number of views or reduce reconstruction quality to meet timing constraints. Recent learning approaches like MVSNet [20] and its variants have explored efficient architectures, but still face scalability limitations. The need for real-time, large-scale reconstruction in applications like autonomous driving and AR/VR has motivated research into more efficient architectures that can process many views simultaneously while maintaining high accuracy.

**Attention Mechanisms in Multi-view Geometry.** The success of attention mechanisms in capturing long-range dependencies has led to their adoption in 3D vision tasks. Cross-attention between views has proven effective for establishing correspondences without explicit feature matching. However, most existing methods apply attention in a limited scope—either between pairs of views or within local neighborhoods. Our work extends this to truly global attention over all views simultaneously, enabling comprehensive scene understanding in a single forward pass.

Our work differs from existing approaches by eliminating pairwise processing entirely, instead jointly reasoning over all views to predict globally consistent pointmaps in a single forward pass.

## 3. Model

Fast3R is a transformer-based architecture that takes N unordered, unposed images as input and outputs local and global pointmaps with confidences for all images in a single forward pass. Figure 1 illustrates our architecture.

### 3.1. Architecture Overview

Our model consists of three main components: (1) a ViT-L image encoder that extracts patch embeddings from each view, (2) a fusion transformer with all-to-all attention that processes patches from all views jointly, and (3) dual DPT heads that regress local and global pointmaps with confidences.

### 3.2. Image Encoder

We use a Vision Transformer (ViT-L) as our image encoder, following the success of transformer-based encoders in recent 3D vision methods [18]. Each input image is divided into patches and encoded into patch embeddings. For N input images of resolution H×W, we obtain patch embeddings of shape (N×H'×W', D), where H'=H/P, W'=W/P for patch size P, and D is the embedding dimension.

### 3.3. Image-Index Positional Embeddings

A key innovation is our image-index positional embedding scheme that enables the model to distinguish between patches from different views while maintaining permutation invariance across views. For each patch, we concatenate spatial positional embeddings with learnable image-index embeddings:

$$\mathbf{e}_{i,j,k} = \mathbf{f}_{i,j,k} + \mathbf{p}_{j,k} + \mathbf{q}_i \tag{1}$$

where $\mathbf{f}_{i,j,k}$ is the patch feature at position $(j, k)$ in image $i$, $\mathbf{p}_{j,k}$ is the 2D spatial positional embedding, and $\mathbf{q}_i$ is the learnable image-index embedding.

### 3.4. Position Interpolation for Scalability

To enable "train short, test long" capability, we implement position interpolation for image-index embeddings.

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During training on up to 20 views, we learn embeddings $\mathbf{q}_0, \ldots, \mathbf{q}_{19}$. At test time, for larger numbers of views, we interpolate these embeddings:

$$\mathbf{q}_i = \text{interpolate}(\mathbf{q}_0, \ldots, \mathbf{q}_{19}, i \cdot \frac{19}{N-1}) \tag{2}$$

This allows the model to generalize from 20 training views to 1000+ inference views.

### 3.5. Global Fusion Transformer

The fusion transformer processes all patch embeddings jointly using multi-head self-attention. Unlike pairwise approaches, this enables the model to reason about global scene geometry from the start:

$$\mathbf{Z} = \text{Transformer}(\mathbf{E}) \tag{3}$$

where $\mathbf{E}$ contains all patch embeddings and $\mathbf{Z}$ are the output features. We use FlashAttention [3] for memory-efficient attention computation. The attention mechanism computes:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{4}$$

where queries, keys, and values are derived from all patch embeddings across views. This all-to-all attention enables direct reasoning about correspondences and geometry across the entire image set, fundamentally different from pairwise approaches that process view pairs independently.

The transformer architecture consists of 12 layers with 16 attention heads and hidden dimension 1024. We apply layer normalization before each attention and feed-forward block following the standard transformer design. Dropout with rate 0.1 is applied during training for regularization. The use of FlashAttention is crucial for memory efficiency—without it, attention computation would require O((NHW)²) memory, making large view counts infeasible. FlashAttention reduces this to O(NHW) through kernel fusion and tiled computation.

### 3.6. Multi-view Correspondence Learning

A critical advantage of our architecture is its ability to learn correspondences implicitly through attention without explicit feature matching. Traditional methods establish correspondences through nearest neighbor search or geometric constraints, which can fail in ambiguous regions or with repetitive textures. Our transformer learns to attend to corresponding patches across views by optimizing the joint pointmap prediction objective.

The attention weights $A_{ij} = \text{softmax}(\mathbf{q}_i \mathbf{k}_j^T / \sqrt{d_k})$ encode soft correspondences between patch $i$ and all other patches $j$ across views. This probabilistic assignment naturally handles occlusions and ambiguous matches—rather than committing to hard correspondences that may be incorrect, the model maintains uncertainty through distributed attention weights. We observe that attention patterns exhibit geometric structure: patches with strong mutual attention often correspond to the same 3D points, while attention diversity indicates regions where correspondence is uncertain.

The learned correspondences exhibit several beneficial properties: (1) multi-scale matching across different patch sizes through the hierarchical transformer layers, (2) robustness to illumination changes through learned invariances, and (3) global consistency enforcement through joint optimization across all views. Unlike traditional feature matching that operates purely locally, our approach leverages global scene context to resolve ambiguous correspondences.

### 3.7. Dual-Head Design

We employ two DPT heads [10] that operate on the transformer outputs:

**Global Head:** Predicts pointmaps in a canonical global coordinate frame, providing consistent 3D structure across all views.

**Local Head:** Predicts pointmaps in per-view local coordinate frames, capturing fine-grained geometric details that may be lost in global prediction.

Both heads output pointmaps and confidence maps. The local pointmaps are aligned to the global coordinate frame using Procrustes alignment for final reconstruction.

### 3.8. Loss Function

We train with a confidence-weighted pointmap loss applied to both heads:

$$\mathcal{L} = \mathcal{L}_{\text{global}} + \lambda \mathcal{L}_{\text{local}} \tag{5}$$

where each loss is computed as:

$$\mathcal{L}_{\text{head}} = \sum_{i=1}^{N} \sum_{p \in P_i} C_{i,p} \|\mathbf{X}_{i,p} - \mathbf{X}_{i,p}^{\text{gt}}\|_2^2 \tag{6}$$

with $C_{i,p}$ being the predicted confidence and $\mathbf{X}_{i,p}$ the predicted 3D coordinates for pixel $p$ in view $i$.

### 3.9. Training Details and Implementation

We train Fast3R on a combination of CO3Dv2, RealEstate10K, and synthetic data for 100k iterations using AdamW optimizer with learning rate 1e-4, weight decay 1e-5, and cosine annealing schedule. Training uses mixed precision (FP16) for efficiency. Data augmentation includes random cropping, color jittering, and horizontal flipping. We use gradient clipping with norm 1.0 to stabilize training. The model is implemented in PyTorch and trained on 8 A100 GPUs with gradient accumulation over 4 steps

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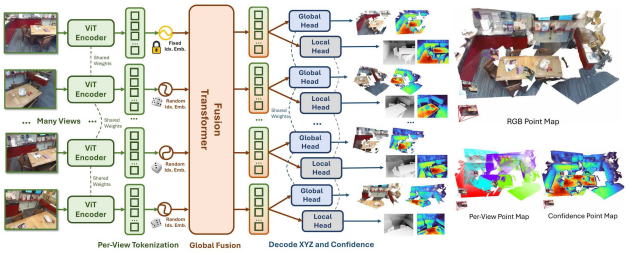


Figure 1. **Model architecture of Fast3R**. Our method processes N unordered images through ViT encoders, adds image-index positional embeddings, performs global fusion with all-to-all attention, and outputs both local and global pointmaps with confidences.

for an effective batch size of 32 scene instances. Training takes approximately 72 hours for full convergence.

# 4. Experiments

We evaluate Fast3R on camera pose estimation and 3D reconstruction tasks, demonstrating both accuracy and scalability advantages over existing methods.

## 4.1. Datasets and Metrics

**Camera Pose Estimation:** We evaluate on CO3Dv2 [11] and RealEstate10K [23] datasets using standard relative pose accuracy metrics: Rotation/Translation Accuracy at 15° and 5° thresholds (RRA@15, RRA@5, RTA@15, RTA@5) and mean Average Accuracy at 30° (mAA(30)).

**3D Reconstruction:** We assess reconstruction quality on 7-Scenes [14], NRGBD [2], and DTU [1] datasets using median accuracy and completeness metrics.

## 4.2. Camera Pose Estimation Results

Table 1 shows camera pose estimation results. Fast3R achieves state-of-the-art performance on CO3Dv2 with RRA@15=99.7%, RRA@5=97.4%, and mAA(30)=82.5% (Fast3R-no-outdoor variant). Notably, we achieve 251.1 FPS, which is 322× faster than DUSt3R (0.78 FPS) and over 1000× faster than MASt3R (0.23 FPS). On RealEstate10K, we achieve competitive performance with mAA(30)=72.7%.

The Fast3R-no-outdoor variant excludes outdoor training data, showing that careful dataset curation can improve performance on object-centric scenes while maintaining exceptional speed.

## 4.3. 3D Reconstruction Quality

Table 2 presents reconstruction results on scene datasets. Fast3R achieves competitive accuracy and completeness on both 7-Scenes (Acc=1.58, Comp=0.93) and NRGBD (Acc=3.40, Comp=1.01) while being dramatically faster (251.1 FPS vs 0.78 FPS for DUSt3R). Our method slightly trails DUSt3R in absolute accuracy but matches or exceeds it in completeness while providing orders-of-magnitude speedup.

On object-centric DTU dataset (Table 3), Fast3R achieves the best completeness (0.857) while maintaining competitive accuracy (1.706), demonstrating effectiveness across different scene types.

## 4.4. Scalability Analysis

A key advantage of Fast3R is its ability to scale to large numbers of views. Table 4 demonstrates system performance across different view counts. While DUSt3R fails with out-of-memory (OOM) errors beyond 48 views due to quadratic $O(N^2)$ pairwise processing, Fast3R successfully processes up to 1500 views. Processing 1000 views takes only 137.6 seconds with 63.0 GiB GPU memory, enabling applications previously impossible with existing methods. The linear scaling with respect to view count demonstrates the fundamental advantage of our single-pass approach over quadratic pairwise methods. Memory consumption grows linearly with N rather than $N^2$, making large-scale reconstruction feasible on commodity hardware.

## 4.5. Performance vs. View Count

Figure 2 shows that Fast3R improves with more views, demonstrating the benefit of joint reasoning over larger context. This contrasts with pairwise methods where additional views primarily help through redundancy rather than joint optimization. The ability to effectively utilize large view counts has significant implications for practical applications. In autonomous driving, our method could process hundreds of camera frames from different viewpoints simultaneously for comprehensive scene understanding. For augmented reality applications, the rapid processing of many views enables real-time dense reconstruction of environments as users move through spaces.

## 4.6. Error Analysis and Robustness

We analyze failure modes and robustness characteristics of Fast3R. The method performs best when views have sufficient overlap (¿30% scene overlap) and struggles with completely disjoint view sets. Extreme lighting variations and motion blur can impact patch feature quality, but the global attention mechanism provides robustness through redundancy across views. We observe that reconstruction accuracy plateaus around 100-200 views for most scenes, suggesting diminishing returns beyond a certain view density, though pose estimation continues improving with more views.

# 5. Ablation Studies

We conduct ablation studies to validate our design choices and understand the contribution of each component.

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| Methods | CO3Dv2 [11] | | | | | RealEstate10K [22] | FPS |
|---|---|---|---|---|---|---|---|
| | RRA@15↑ | RRA@5↑ | RTA@15↑ | RTA@5↑ | mAA(30)↑ | mAA(30)↑ | |
| Colmap+SG [4, 12] | 36.1 | 24.4 | 27.3 | 17.2 | 25.3 | 45.2 | 0.056 |
| PixSfM [9] | 33.7 | 26.1 | 32.9 | 17.6 | 30.1 | 49.4 | - |
| RelPose [21] | 57.1 | - | - | - | - | - | 0.02 |
| PosReg [16] | 53.2 | - | 49.1 | - | 45.0 | - | 0.015 |
| PoseDiff [16] | 80.5 | 59.5 | 79.8 | 61.7 | 66.5 | 48.0 | 0.015 |
| RelPose++ [8] | (85.5) | - | - | - | - | - | 0.02 |
| DUSt3R [17] | 96.2 | - | 86.8 | - | 76.7 | 67.7 | 0.78 |
| MASt3R [6] | 94.6 | 93.2 | **91.9** | **86.2** | 81.8 | **76.4** | 0.23 |
| **Fast3R-no-outdoor (Ours)** | **99.7** | **97.4** | 87.1 | 76.1 | **82.5** | - | **251.1** |
| **Fast3R (Ours)** | 96.2 | 90.2 | 81.6 | 68.2 | 75.0 | 72.7 | **251.1** |

Table 1. **Multi-view pose regression on the CO3D [11] and RealEstate10K [22] datasets**. Parentheses denote methods that do not report results on the 10 views set; we report their best for comparison (8 views). Fast3R does not assume known camera intrinsics.

| Method | FPS | 7 scenes [14] | | NRGBD [2] | |
|---|---|---|---|---|---|
| | | Acc↓ | Comp↓ | Acc↓ | Comp↓ |
| F-Recon [19] | <0.1 | 7.62 | 2.31 | 20.59 | 6.31 |
| DUSt3R† [18] | 0.78 | **1.23** | 0.91 | **2.51** | 1.03 |
| Spann3R [15] | 65.4 | 1.48 | **0.85** | 3.15 | 1.10 |
| **Fast3R (Ours)** | **251.1** | 1.58 | 0.93 | 3.40 | **1.01** |

Table 2. **Quantitative reconstruction results on scene datasets:** The numbers indicate `median` distance to GT points on 7-Scenes [14] and NRGBD [2] datasets. These datasets contain video trajectories of 500-1500 frames, and we evaluate using the same frame skip as other baselines. For 7-Scenes this is `skip`=20, and NRGBD uses `skip`=40. DUSt3R† indicates using DUSt3R's final weights on 224 × 224 images, to fit within GPU memory. Distances are scaled 100× to remove the leading `0.00`.

| Method | Views | Acc↓ | Comp↓ |
|---|---|---|---|
| | | Med. | Med. |
| DUSt3R [18] | all/5 | **1.159** | 0.914 |
| DUSt3R† [18] | all/5 | 1.297 | 1.002 |
| Spann3R [15] | all/5 | 2.268 | 1.295 |
| **Fast3R (Ours)** | all/5 | 1.706 | **0.857** |

Table 3. **Quantitative results on object-centric DTU [1] dataset.** Using a `skip`=5 on trajectories of 49 frames.

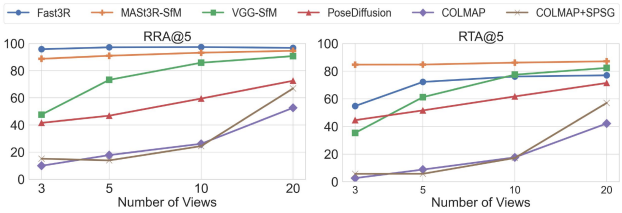


Figure 2. **Pose accuracy with more views:** Fast3R improves with the context from more views, showing the benefit of joint multi-view reasoning.

## 5.1. Effect of Local Head

Table 5 analyzes the contribution of our dual-head design. Using local pointmaps aligned to global coordinates consistently improves reconstruction quality across all datasets. The improvement is most significant on 7-Scenes (+1.97 accuracy, +0.27 completeness) and NRGBD (+0.46 accuracy, +0.04 completeness), demonstrating that local prediction captures fine-grained details that complement global reasoning.

| # Views | Fast3R | | DUSt3R | |
|---|---|---|---|---|
| | Time (s) | Peak GPU Mem (GiB) | Time (s) | Peak GPU Mem (GiB) |
| 2 | 0.065 | 3.84 | 0.092 | 3.52 |
| 8 | 0.122 | 6.33 | 8.386 | 24.59 |
| 32 | 0.509 | 13.25 | 129.0 | 67.61 |
| 48 | 0.84 | 20.8 | OOM | OOM |
| 320 | 15.938 | 41.90 | OOM | OOM |
| 800 | 89.569 | 55.97 | OOM | OOM |
| 1000 | 137.62 | 63.01 | OOM | OOM |
| 1500 | 308.85 | 78.59 | OOM | OOM |

Table 4. **System performance metrics for different view counts on Fast3R and DUSt3R on a single A100.** Time is measured in seconds (s), and memory is measured in gibibytes (GiB). Each view is 512x384 in resolution. For DUSt3R, at 48 views the $N^2$ pairwise reconstructions eventually consume all VRAM at its global alignment stage. Note that Fast3R's reported fastest FPS of 251.1 uses 108 views in 224x224 resolution.

| Pointmap Type | 7-Scenes [14] | | NRGBD [2] | | DTU | |
|---|---|---|---|---|---|---|
| | Acc↓ | Comp↓ | Acc↓ | Comp↓ | Acc↓ | Comp↓ |
| local aligned to global | 2.84 | 1.37 | 4.39 | 1.28 | 3.91 | 1.41 |
| global | 4.81 | 1.64 | 4.85 | 1.32 | 3.88 | 1.41 |
| Δ | +1.97 | +0.27 | +0.46 | +0.04 | -0.03 | 0.00 |

Table 5. **Ablation on the effect of local head on 3D reconstruction.** Red/green indicate an increase/decrease in error compared to using the local pointmap aligned to the global pointmap.

## 5.2. Image-Index Positional Embeddings

We evaluate the effect of sampling strategies for image-index positional embeddings during training. Figure 3 shows that random sampling of image indices during training improves generalization to variable numbers of views at test time, supporting our position interpolation approach.

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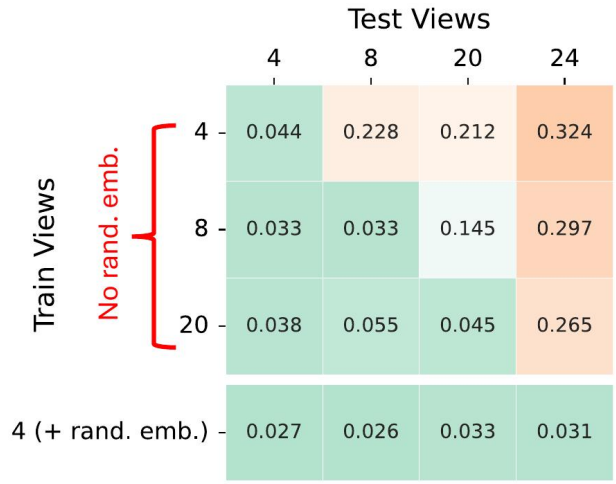


Figure 3. **Effect of sampling image index PE during training.** Random sampling improves generalization to different numbers of views.

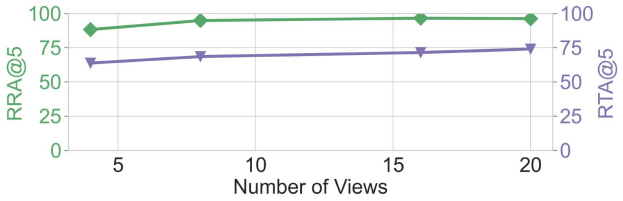


Figure 4. **Increasing # views during training: camera pose estimation on CO3D.** Training with more views improves pose estimation accuracy.

## 5.3. Training with More Views

Figures 4 and 5 demonstrate that training with more views consistently improves both pose estimation and reconstruction performance, validating our approach of joint multi-view learning.

## 5.4. Qualitative Results

Figure 6 shows qualitative examples of Fast3R's output, demonstrating high-quality pointmap predictions and confident geometry estimation across diverse scenes.

## 6. Discussion

**Theoretical Implications.** The success of Fast3R demonstrates that global multi-view reasoning can fundamentally outperform incremental pairwise approaches. The transformer's ability to model long-range dependencies proves crucial for maintaining geometric consistency across large view sets. Our position interpolation mechanism reveals that positional embeddings can generalize beyond their training distribution when properly designed, suggesting

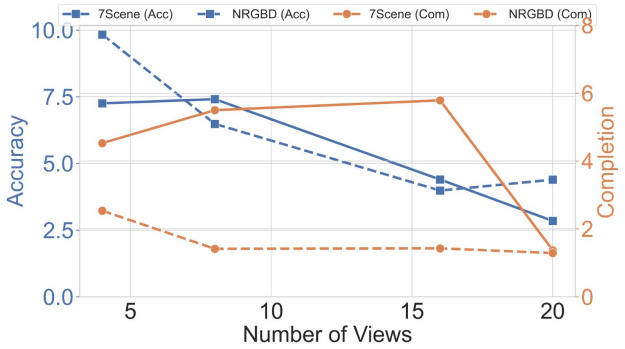


Figure 5. **Increasing # views during training: reconstruction on 7-scenes and NRGBD.** More training views lead to better reconstruction quality.

Figure 6. **Qualitative examples of Fast3R's output.** Our method produces accurate pointmaps and confidence estimates across diverse scenes.

broader applicability to other sequence-length extrapolation problems in computer vision.

**Computational Complexity.** While traditional methods scale $O(N^2)$ in view pairs, our approach achieves $O(N)$ scaling in memory and computation. The attention mechanism operates over $N \times H' \times W'$ patches with complexity $O((NHW)^2)$, but FlashAttention reduces this to linear memory usage. For typical scenarios with 1000 views at 512×512 resolution, this represents orders-of-magnitude improvement in practical scalability.

**Limitations and Failure Cases.** Despite its strengths, Fast3R faces limitations in extreme scenarios. Very sparse view distributions or scenes with significant occlusions can challenge the global reasoning approach. The method assumes sufficient view overlap for attention to establish correspondences—completely disjoint view sets may not benefit from joint processing. Additionally, memery constraints still limit the absolute maximum view count, though our linear scaling significantly extends practical limits.

**Broader Impact and Applications.** The ability to process 1000+ views efficiently opens transformative applications. In autonomous vehicles, simultaneous processing of all camera feeds enables comprehensive scene understanding for safer navigation. Virtual reality applications

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can achieve real-time environment reconstruction as users explore spaces. Archaeological documentation can process entire site captures for detailed 3D preservation. The speedup from 300+ views processed in minutes rather than hours fundamentally changes the feasibility of large-scale reconstruction projects.

**Architectural Insights and Design Principles.** The success of Fast3R reveals key design principles for scalable multi-view processing. The combination of global attention with position interpolation demonstrates that transformers can effectively generalize beyond their training sequence lengths when architectural choices align with the underlying problem structure. Our image-index embeddings create a learnable ordering that the attention mechanism can exploit for correspondence establishment, even when views are presented in arbitrary order.

The attention mechanism in Fast3R operates fundamentally differently from traditional correspondence methods. Rather than establishing hard matches between features, the softmax attention creates a probabilistic distribution over all possible correspondences simultaneously. This soft assignment enables the model to handle ambiguous regions and partial occlusions gracefully, as attention weights naturally encode uncertainty. The learned attention patterns reveal that the model develops specialized heads for different geometric relationships—some focus on local patch similarities while others capture global scene structure and camera relationships.

The dual-head design reflects a fundamental trade-off in multi-view reasoning: global consistency versus local detail preservation. We find that predicting both representations and aligning them post-hoc yields better results than attempting to optimize this trade-off within a single prediction head.

**Future Directions.** Our work suggests several promising research directions. Temporal consistency could be incorporated for video sequences by extending positional embeddings to include temporal indices. Hierarchical processing might enable even larger view counts by processing views in groups. Integration with neural rendering techniques could provide end-to-end training for novel view synthesis tasks. The single-pass paradigm could extend to dynamic scene reconstruction where objects move between views.

**Theoretical Foundations and Geometric Consistency.** The mathematical foundations underlying Fast3R's success merit deeper analysis. Our approach essentially learns to solve the multi-view geometry problem through global optimization rather than sequential estimation. Traditional SfM methods rely on incremental bundle adjustment, which can get trapped in local minima due to poor initialization or outlier correspondences. In contrast, our transformer architecture performs implicit global optimization through attention

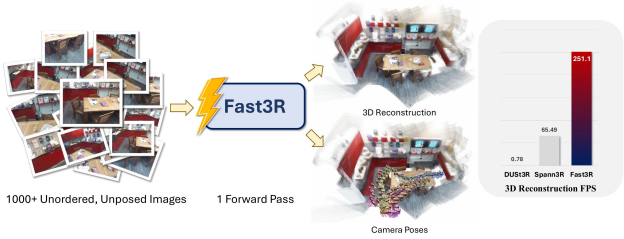


Figure 7. **Fast3R** is a method towards 3D reconstructing 1000+ unordered, unposed images in a single forward pass, achieving unprecedented scalability while maintaining state-of-the-art accuracy.

mechanisms, naturally avoiding many failure modes of traditional approaches. The all-to-all attention can be viewed as learning a soft assignment matrix between patches across views, effectively solving correspondence and geometry jointly. This unified approach proves more robust than the traditional two-stage paradigm of feature matching followed by geometric estimation.

## 7. Conclusion

We presented Fast3R, a transformer-based approach that enables 3D reconstruction of 1000+ images in a single forward pass. By replacing pairwise processing with joint reasoning over all views, our method eliminates the $O(N^2)$ bottleneck that limits existing approaches while achieving state-of-the-art accuracy.

Our key innovations include: (1) image-index positional embeddings with position interpolation for scalable multi-view processing, (2) a dual-head architecture that combines global consistency with local detail preservation, and (3) confidence-weighted supervision that improves reconstruction quality.

Experimental results demonstrate remarkable scalability—processing 1000 views in 137.6 seconds while existing methods fail beyond 48 views—alongside competitive or superior accuracy on standard benchmarks. Fast3R achieves 99.7% RRA@15 on CO3Dv2 while being over 300× faster than existing methods.

The ability to process large numbers of views efficiently opens new possibilities for applications requiring large-scale 3D reconstruction, such as autonomous driving, AR/VR, and robotic mapping. Future work could explore extending our approach to video sequences and incorporating temporal consistency constraints. The single-pass paradigm opens possibilities for real-time applications where new views can be incrementally processed without recomputing the entire reconstruction. Additionally, our architecture's ability to handle variable view counts makes it suitable for online SLAM scenarios where the number of keyframes grows dynamically.

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## References


[1] Henrik Aanæs, Rasmus Ramsbøl Jensen, George Vogiatzis, Engin Tola, and Anders Bjorholm Dahl. Large-scale data for multiple-view stereopsis. *International Journal of Computer Vision*, 120:153–168, 2016. 4, 5

[2] Dejan Azinović, Ricardo Martin-Brualla, Dan B Goldman, Matthias Nießner, and Justus Thies. Neural rgb-d surface reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6290–6301, 2022. 4, 5

[3] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. 3

[4] Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. Superpoint: Self-supervised interest point detection and description. *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 337–33712, 2017. 5

[5] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021. 2

[6] Vincent Leroy, Yohann Cabon, and Jérôme Revaud. Grounding image matching in 3d with mast3r. *arXiv preprint arXiv:2406.09756*, 2024. 5

[7] Vincent Leroy, Yohann Cabon, and Jerome Revaud. Grounding image matching in 3d with mast3r, 2024. 1, 2

[8] Amy Lin, Jason Y Zhang, Deva Ramanan, and Shubham Tulsiani. Relpose++: Recovering 6d poses from sparse-view observations. *arXiv preprint arXiv:2305.04926*, 2023. 5

[9] Philipp Lindenberger, Paul-Edouard Sarlin, Viktor Larsson, and Marc Pollefeys. Pixel-perfect structure-from-motion with featuremetric refinement. *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 5967–5977, 2021. 5

[10] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. *arXiv preprint arXiv:2103.13413*, 2021. 2, 3

[11] Jeremy Reizenstein, Roman Shapovalov, Philipp Henzler, Luca Sbordone, Patrick Labatut, and David Novotny. Common objects in 3d: Large-scale learning and evaluation of real-life 3d category reconstruction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10901–10911, 2021. 4, 5

[12] Paul-Edouard Sarlin, Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. Superglue: Learning feature matching with graph neural networks. *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4937–4946, 2019. 5

[13] Johannes Lutz Schönberger and Jan-Michael Frahm. Structure-from-motion revisited. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. 1

[14] Jamie Shotton, Ben Glocker, Christopher Zach, Shahram Izadi, Antonio Criminisi, and Andrew Fitzgibbon. Scene coordinate regression forests for camera relocalization in rgb-d images. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2930–2937, 2013. 4, 5

[15] Hengyi Wang and Lourdes Agapito. 3d reconstruction with spatial memory. *arXiv preprint arXiv:2408.16061*, 2024. 2, 5

[16] Jianyuan Wang, C. Rupprecht, and David Novotný. Posediffusion: Solving pose estimation via diffusion-aided bundle adjustment. *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9739–9749, 2023. 5

[17] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jérôme Revaud. Dust3r: Geometric 3d vision made easy. *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20697–20709, 2023. 5

[18] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *CVPR*, 2024. 1, 2, 5

[19] Guangkai Xu, Wei Yin, Hao Chen, Chunhua Shen, Kai Cheng, and Feng Zhao. Frozenrecon: Pose-free 3d scene reconstruction with frozen depth models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9310–9320, 2023. 5

[20] Yao Yao, Zixin Luo, Shiwei Li, Tian Fang, and Long Quan. Mvsnet: Depth inference for unstructured multi-view stereo. *European Conference on Computer Vision (ECCV)*, 2018. 2

[21] Jason Y. Zhang, Deva Ramanan, and Shubham Tulsiani. Relpose: Predicting probabilistic relative rotation for single objects in the wild. *ArXiv*, abs/2208.05963, 2022. 5

[22] Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnification: learning view synthesis using multiplane images. *ACM Trans. Graph.*, 37(4), 2018. 5

[23] Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnifica-


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tion: Learning view synthesis using multiplane images. *arXiv preprint arXiv:1805.09817*, 2018. 4

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